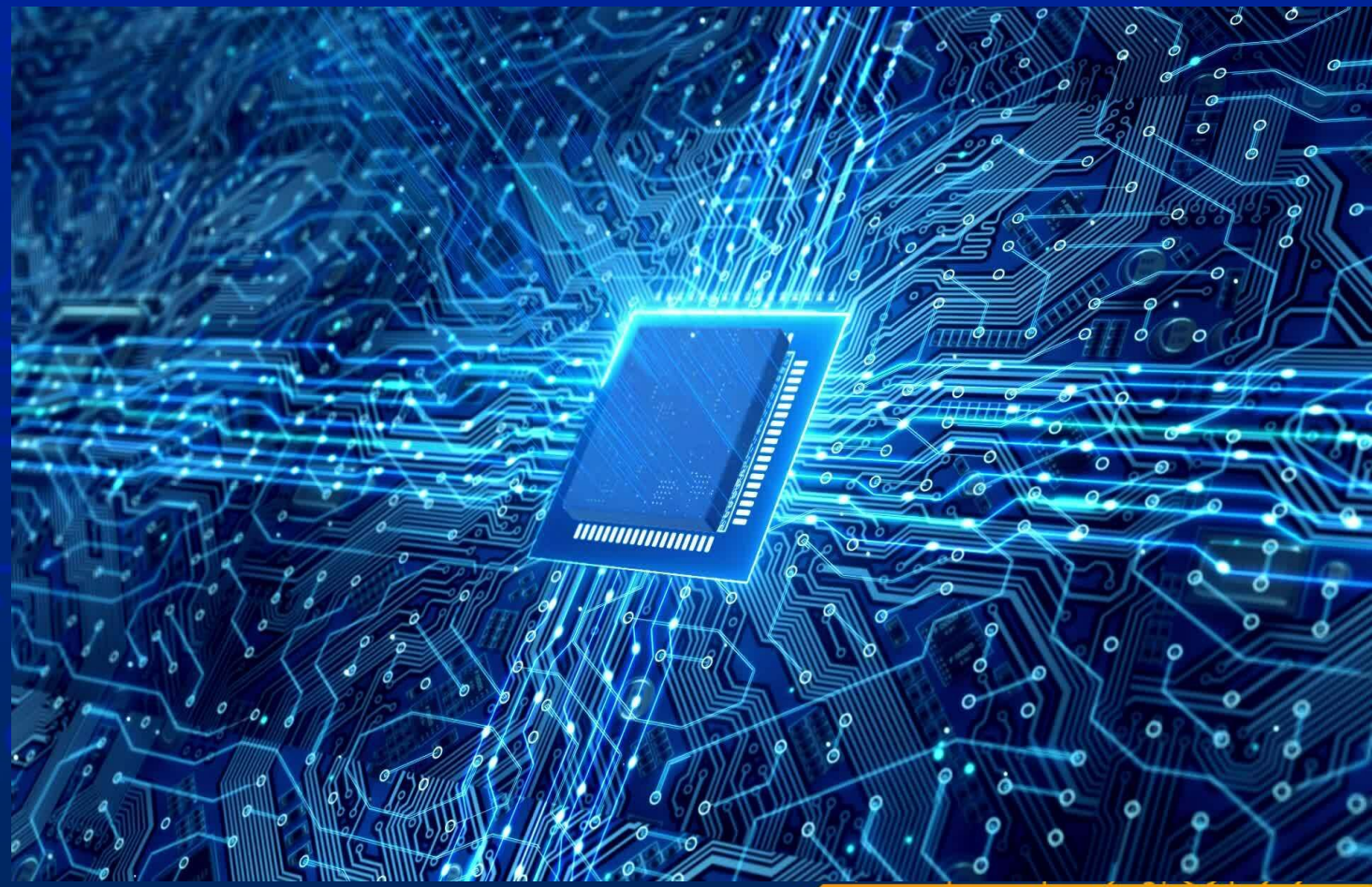
OPEN PROJECT SPACE

Lecture 3:

Integrated Circuits



IEEE

IEEE UF Student Branch

Welcome to Lecture 3

OPEN PROJECT SPACE

- First, let's review debugging
- Then we will cover:
  - Using ICs and pinouts
  - Circuit Schematics
  - Capacitors and Potentiometers
- Project: 555 Piano 🎵🕺

IEEE

IEEE UF Student Branch

Debugging



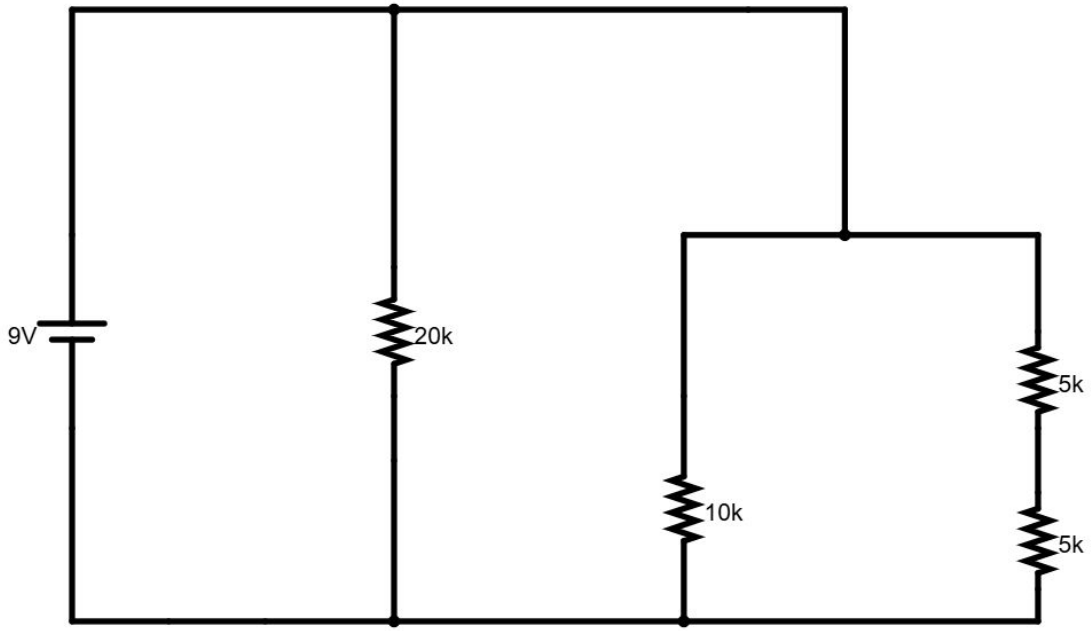
- Debugging is essential knowledge for creating all your future designs!
- Compare expected result to real measurements to find errors
- Allows us to see what's happening within the circuit

Resistance Review

Resistance Review

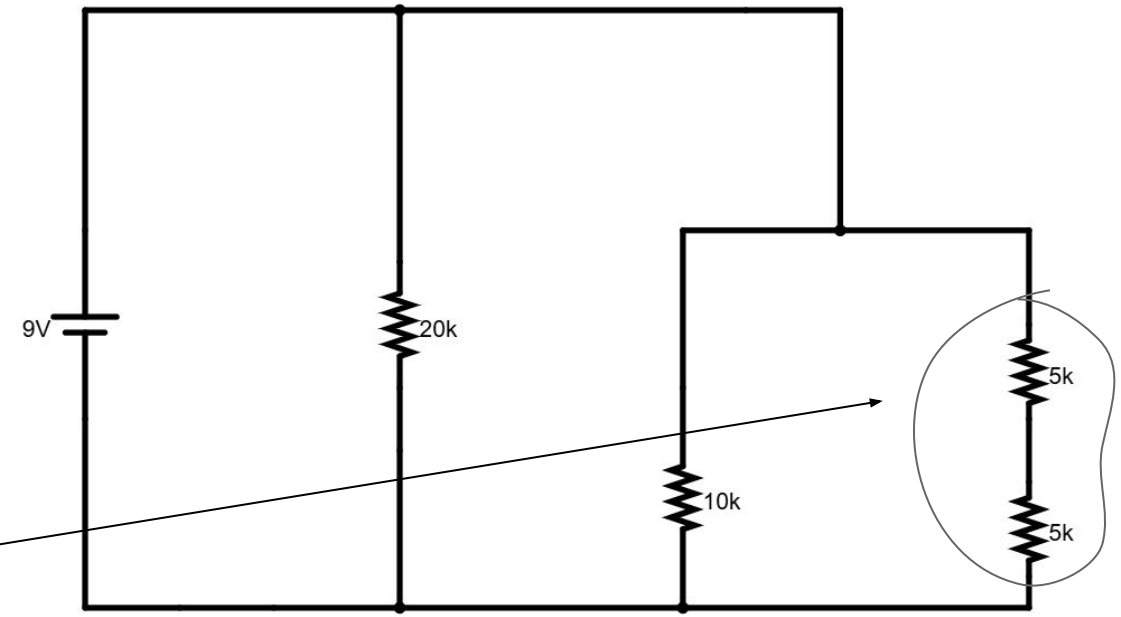
What is the total resistance in this circuit?

Let's break it down



Step by Step Breakdown

First, evaluate series resistance

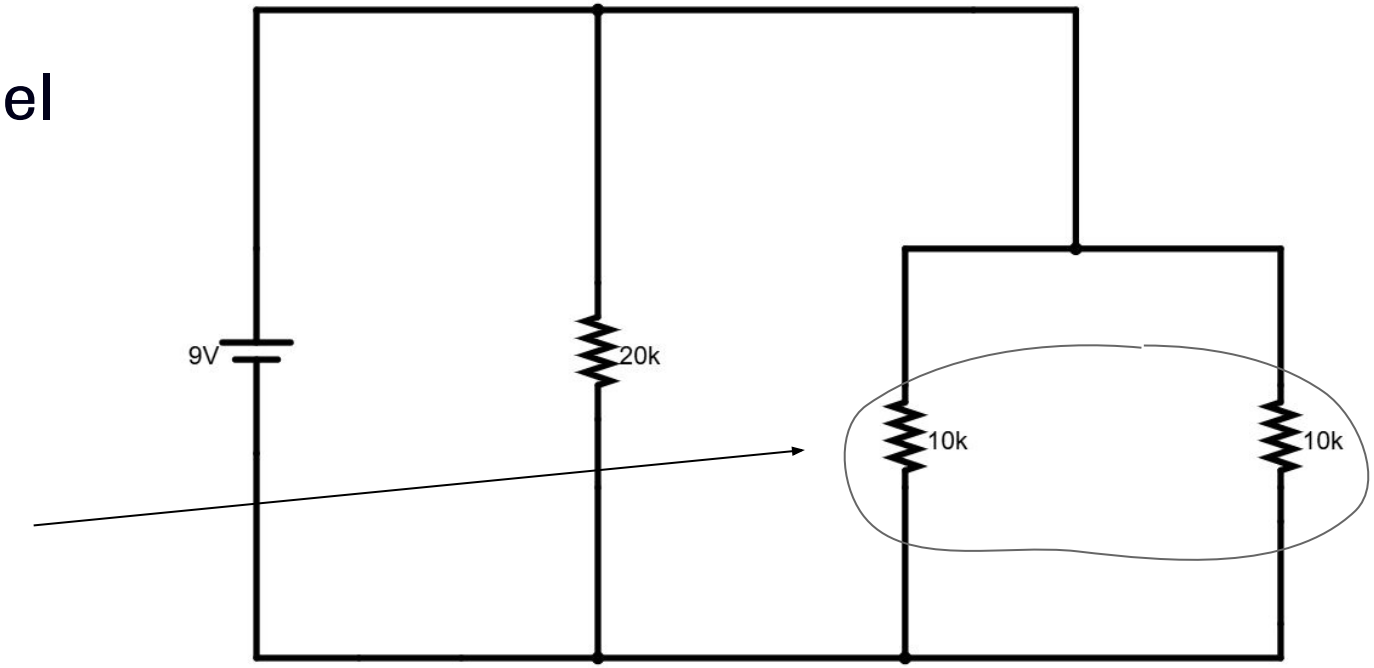


10kΩ

Step by Step Breakdown

Now, evaluate parallel resistance

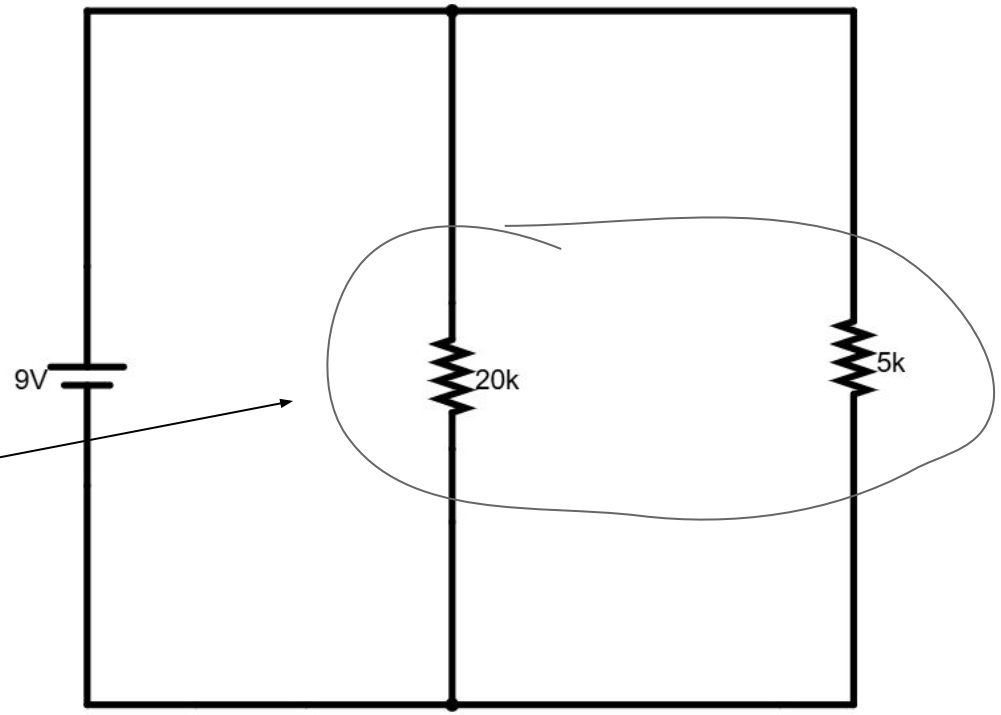
$$\frac{10\ \text{k}\Omega \times 10\ \text{k}\Omega}{10\ \text{k}\Omega + 10\ \text{k}\Omega} = 5\ \text{k}\Omega$$



Step by Step Breakdown

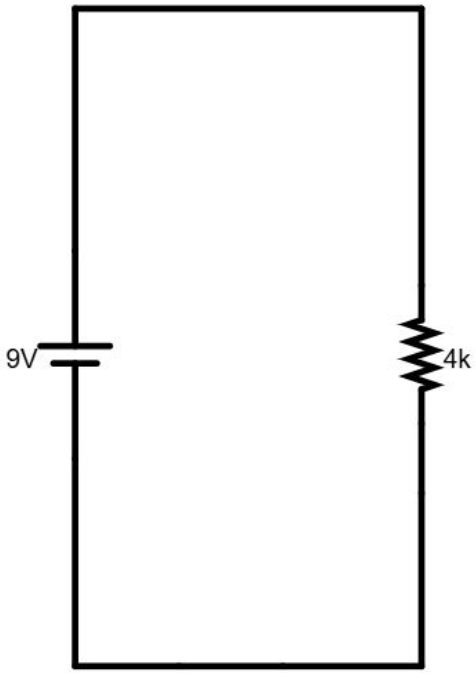
Lastly, evaluate the parallel resistance again

$$\frac{20\ \text{k}\Omega \times 5\ \text{k}\Omega}{20\ \text{k}\Omega + 5\ \text{k}\Omega}$$



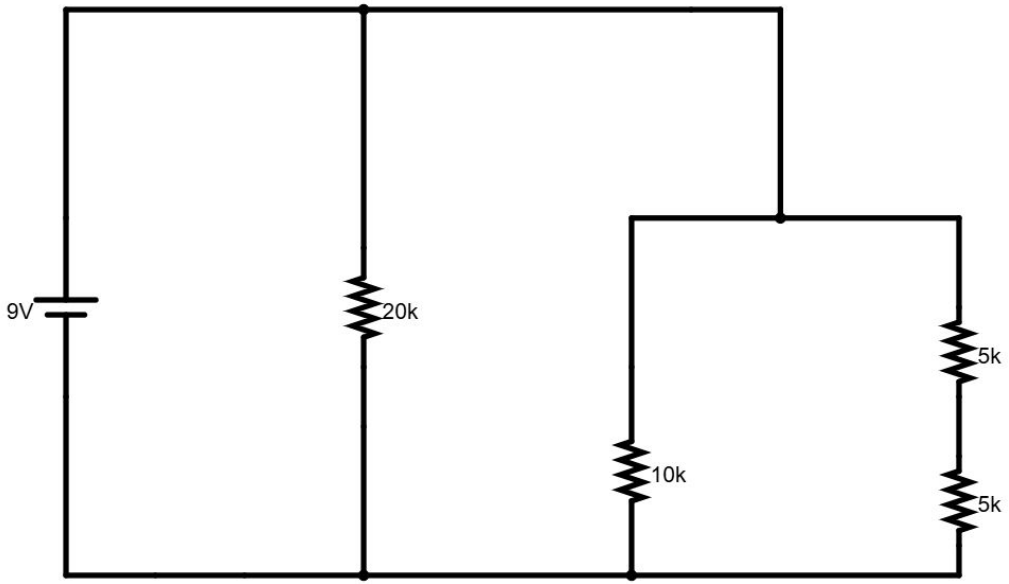
Step by Step Breakdown

Final answer: Req = 4 kΩ



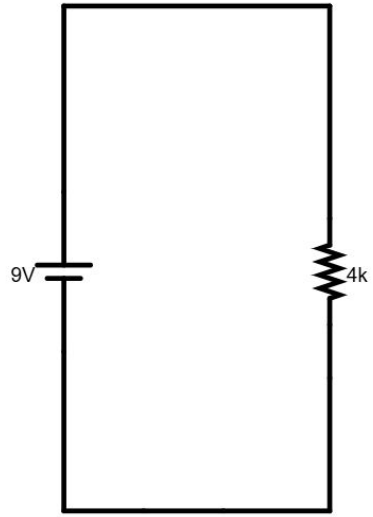
Circuit Analysis!

Original Circuit



Circuit Analysis

Simplified Circuit

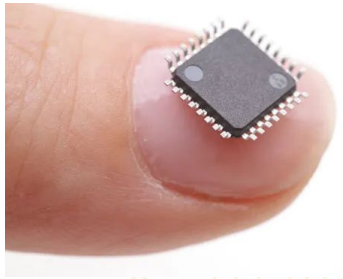


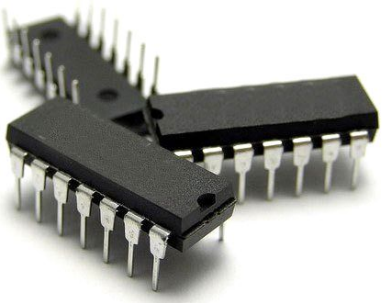
What is an IC?

Simplify Your Board Design

OPEN PROJECT SPACE

- IC's are used to replace discrete circuits
  - Minimize design complexity
  - Cut design costs
  - Save board space
- Composed of resistors, capacitors, transistors, etc.





IEEE

IEEE UF Student Branch

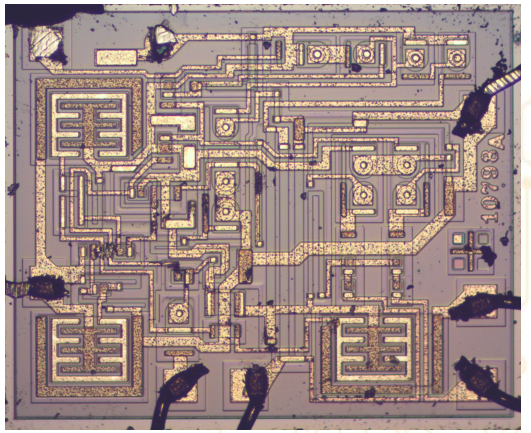
Many applications!

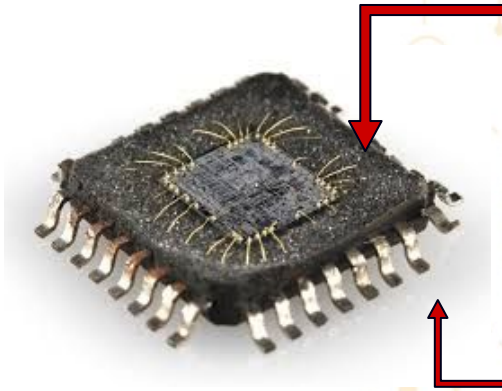
- In your kits you will find:
  - One 555 Precision Timer
  - One 8-bit Shift Register
  - Arduino (ELEGOO) Nano with ATmega328p IC
- IC’s are used in almost all electronics, acting as amplifiers, oscillators, counters, memory, and more!

What's inside?

- IC's contain a die (aka chip)
  - which is connected to external pins
- The die contains the circuit
  - The circuit is made using semiconductor materials

The circuit inside a 555 timer IC

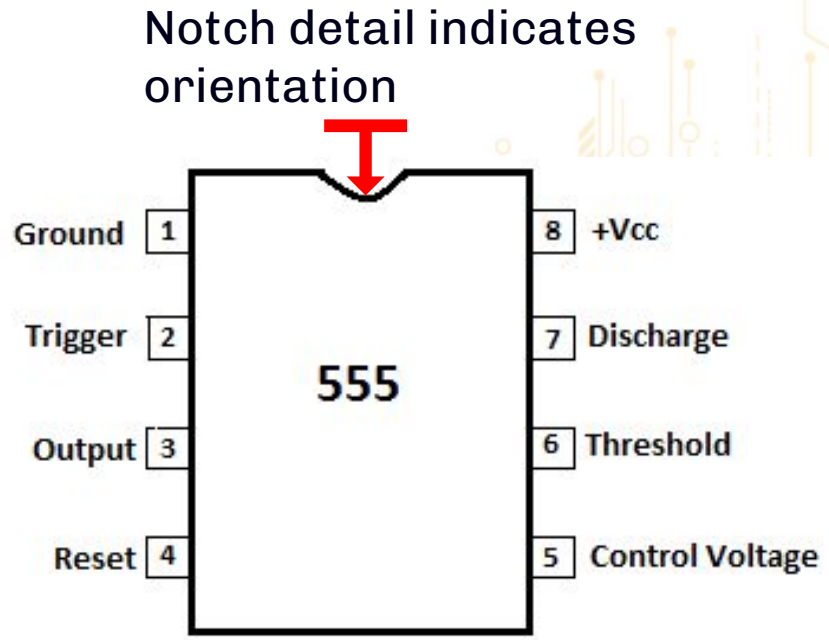




Simplify your Board Design

OPEN PROJECT SPACE

- Pins are how we connect our circuit with the IC
- Pinout diagrams help us understand the purpose of each pin, and gives us info on how to connect it to our circuit



Pins are numbered counterclockwise starting from the top left (top view)

IEEE

IEEE UF Student Branch

OPEN PROJECT SPACE

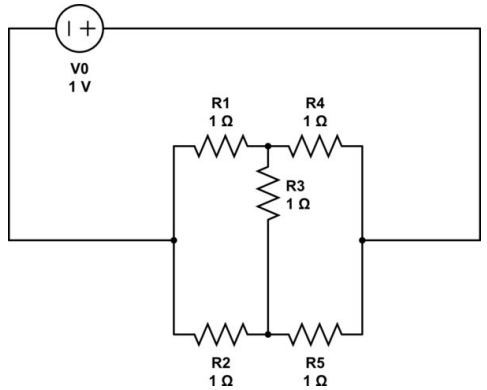
Schematics

Representing Circuits

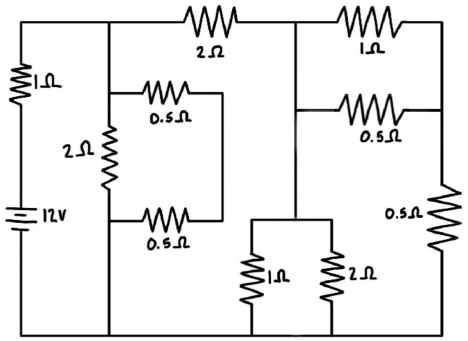
IEEE

IEEE UF Student Branch

- Schematics are organized diagrams of a circuit
  - You will see these all the time (and have already)!
- They use symbols to represent circuit components

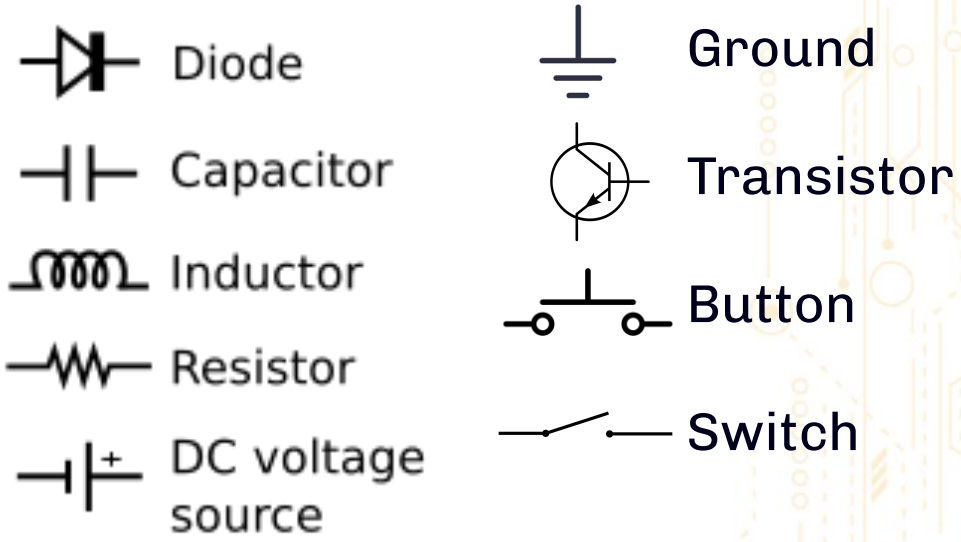


examples from L1



Circuit Symbols

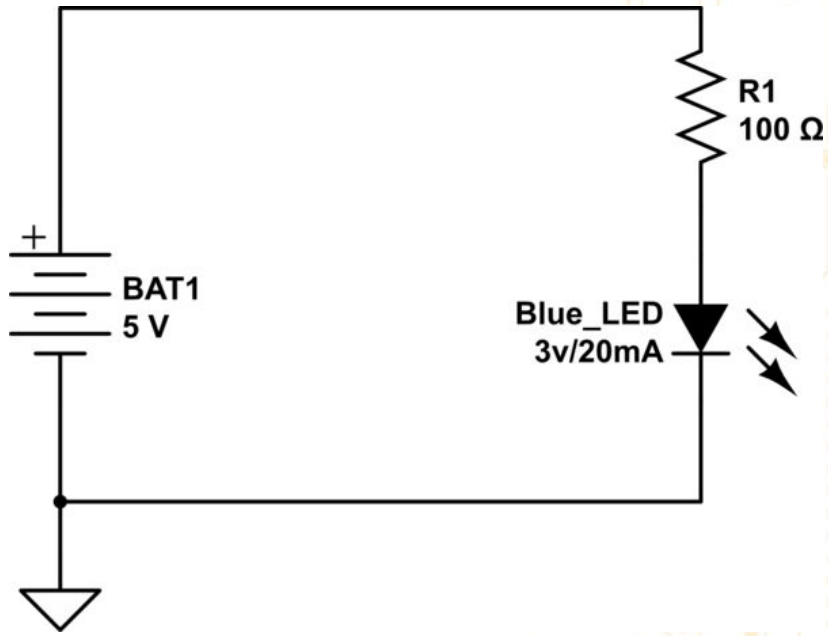
- These are common components you should remember!
  - Symbols may vary based on style



Intro to Reading Schematics

OPEN PROJECT SPACE

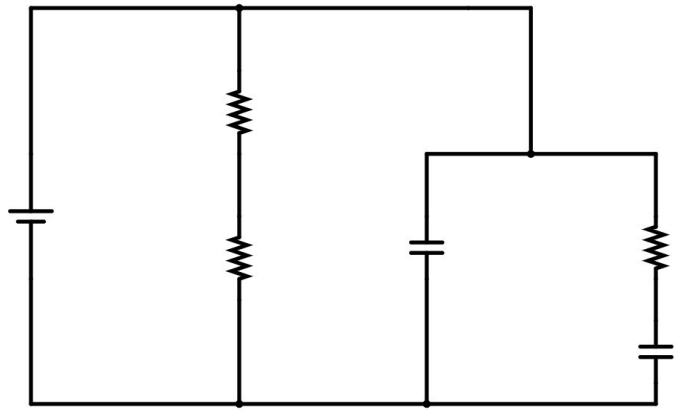
- This is a simple LED circuit schematic, which you know how to build on a breadboard
- Notice the numerical details and variations to symbols
  - A good schematic is easy to understand and fabricate!



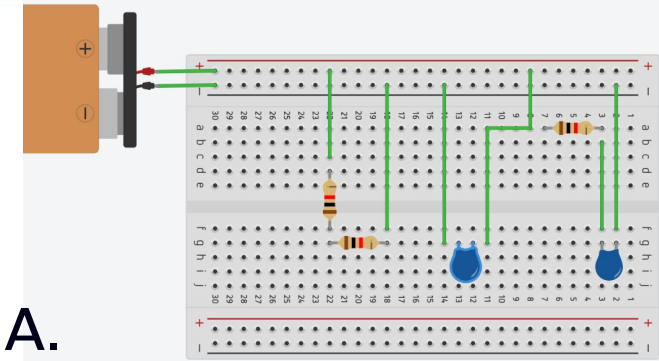
Match the Circuit and Schematic

OPEN PROJECT SPACE

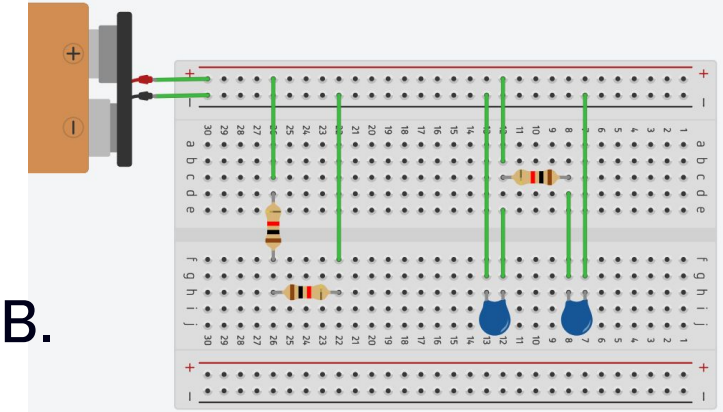
Match the schematic



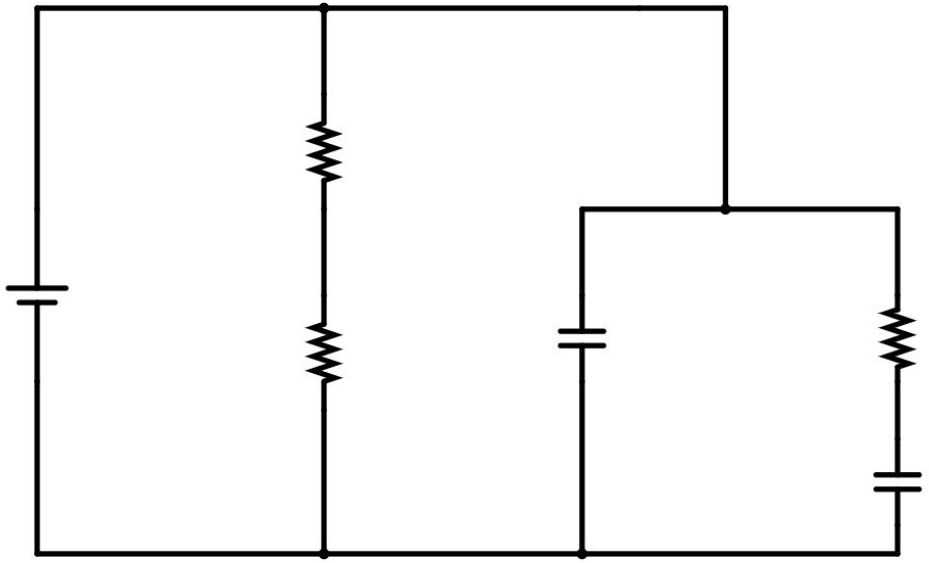
A.

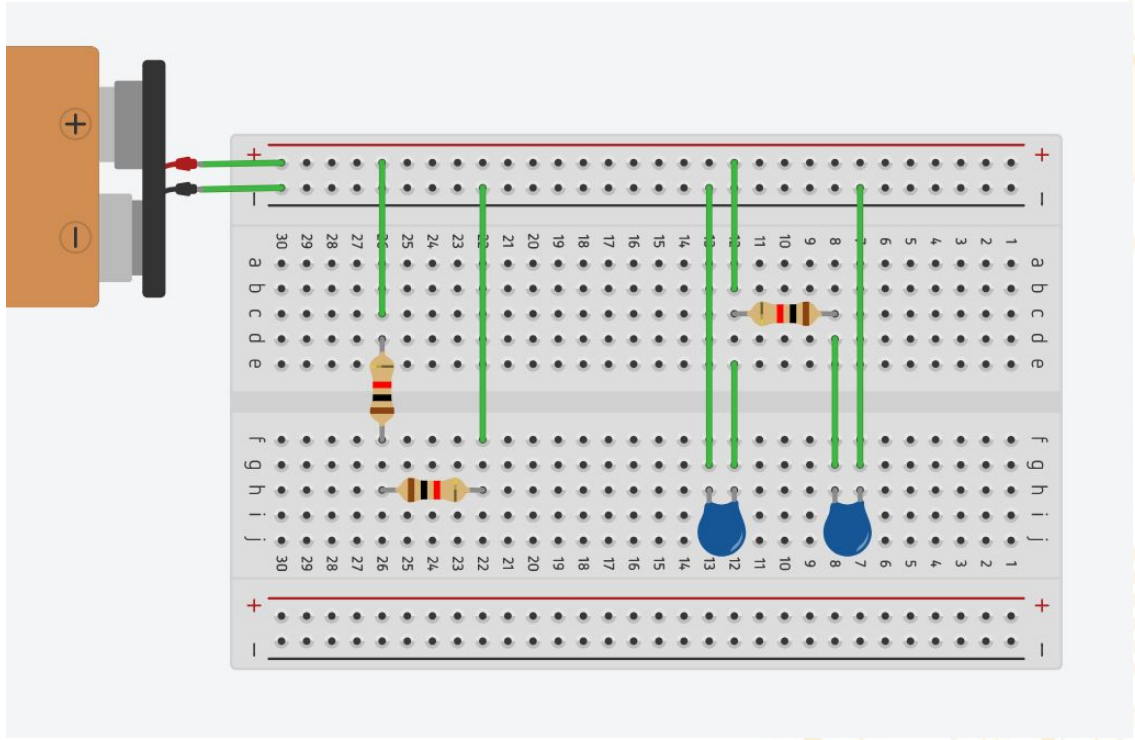


B.



Correct Answer (B)





OPEN PROJECT SPACE

Capacitors

How do they work

IEEE

IEEE UF Student Branch

Capacitors Intro

- Capacitors store electric charge
  - Capacitance represents the amount of charge stored for a given voltage
- Charges and discharges based on properties of the circuit

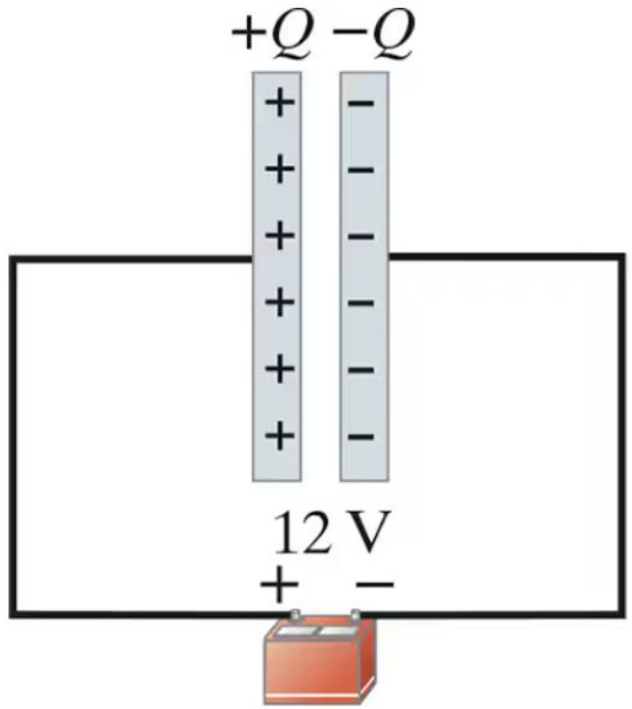


0.1µF ceramic capacitor (in your kits)

Note: ceramic capacitors are non polar, so direction doesn't matter

Inside a Capacitor

- Consists of two parallel plates (may also be rolled)
- Charge equals capacitance times voltage across the plates
  - Q = CV
- C is proportional to plate area over distance between plates
  - C ∝ A/d



Capacitor Formulas

- Capacitors act according to
  - $i = C \times dv/dt$
  - Current equals capacitance times the rate of change of voltage
- Differential equations ->
  - $V(t) = V_{initial} e^{-t/RC}$

current through capacitor

$$i_c = C \frac{dV}{dt}$$

current through resistor

$$i_R = V/R$$

$$i_C + i_R = 0 \leftarrow \text{conservation!}$$

$$C \frac{dV}{dt} + V/R = 0$$

$$\frac{1}{V} \frac{dV}{dt} = -\frac{1}{RC}$$

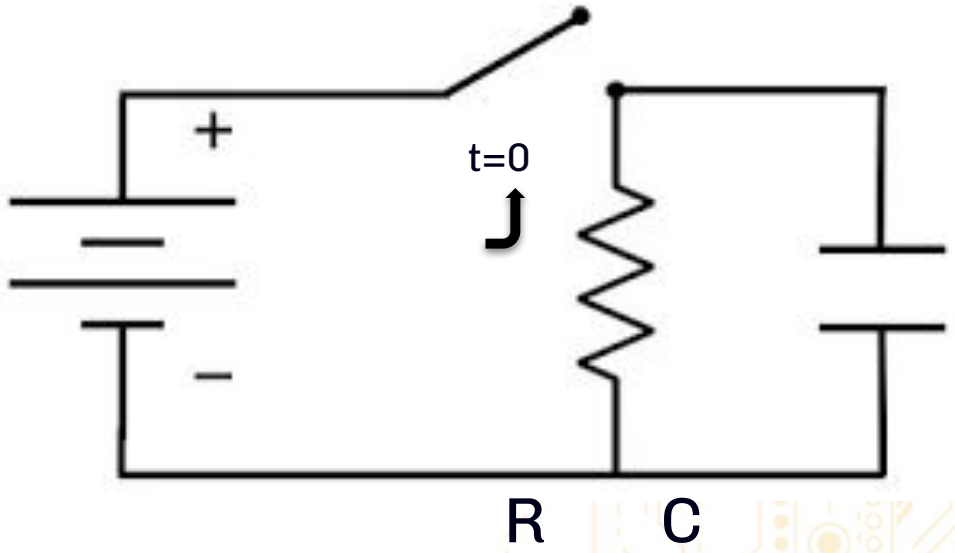
$$\int \frac{1}{V} dV = -\frac{1}{RC} \int dt$$

$$\ln(V) = -\frac{t}{RC} + \text{constant}$$

$$V(t) = V_{initial} e^{-t/RC}$$

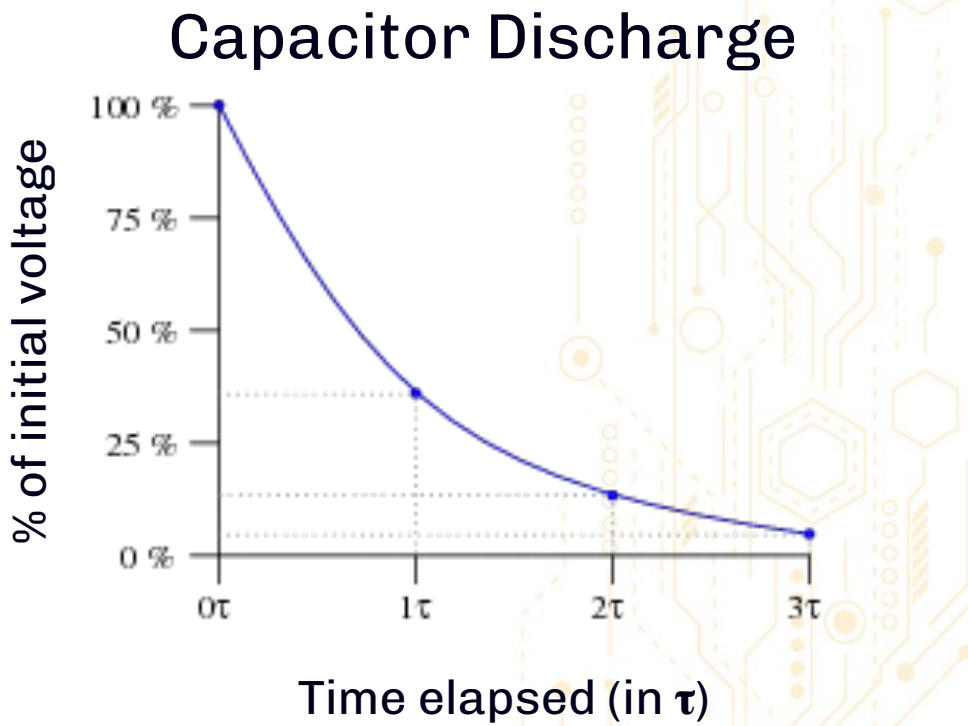
Important Concepts

- **Time constant** of a resistor-capacitor circuit, τ, equals resistance times capacitance (τ=RC)
- One time constant passes -> $V_C = V_{Initial}/e = \sim 0.368 V_{Initial}$



Important Concepts

- A capacitor's voltage discharges according to tau:
- This means that when you **increase resistance** (or capacitance) in the circuit, the capacitor will **discharge/charge more slowly**



Use Case: Hardware Debounce

- When you press a switch, you are connecting two rails
- You may only want them to touch ONCE
- Use a capacitor to prevent instant voltage changes
  - https://www.digikey.com/en/articles/how-to-implement-hardware-debounce-for-switches-and-relays



Introducing the 555 Piano

The Goal

- Using a **555 timer** and a **piezo speaker**, make a “piano” circuit on your breadboard
- Keys could be buttons!



Making Tunes

- A **piezo speaker** works by the piezoelectric effect, material changes shape when electric charge is applied
  - Turns a quickly alternating voltage into **sound waves**
- User input to the circuit should determine the pitch of the speaker (pitch has to do with **frequency**)

Caution: it's polar!



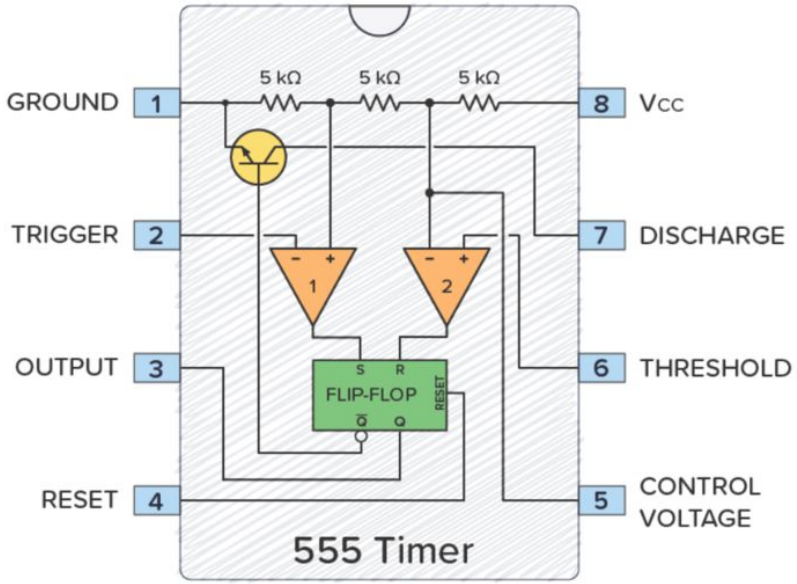
5V magnetic buzzer
(in your kits)

555 Piano

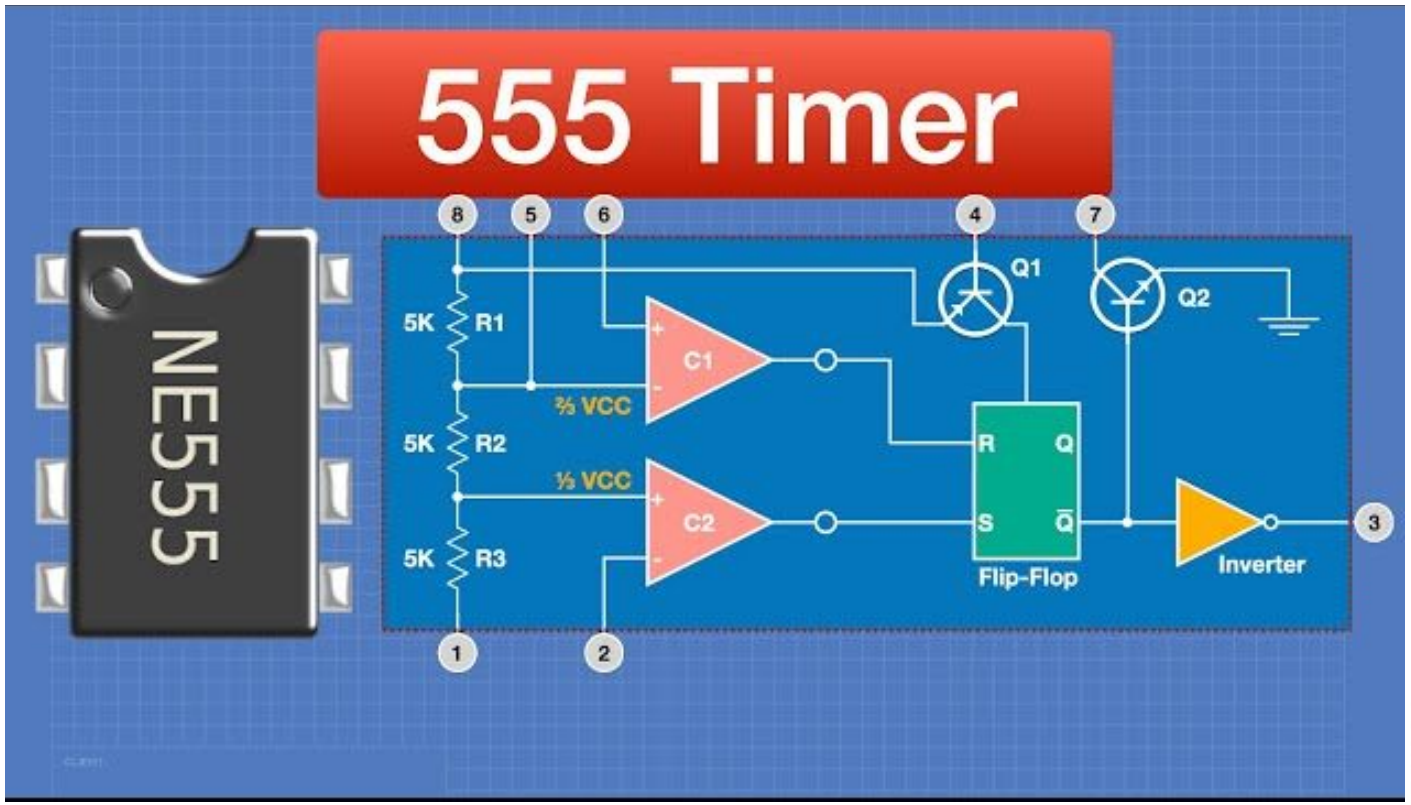
- In this project you use a 555 Timer IC
  https://www.falstad.com/circuit/circuitjs.html
  DigiKey 555 Timer Calculator
- We are using this IC as an **oscillator**, to send out pulses of voltages (low and high) at very specific intervals
- Oscillation speed depends on the **time constant** of the circuit, which we control with inputs to the IC

555 Timer Schematic

- Brief look inside the 555 Timer
  https://tinyurl.com/2c3c4wae
- Each pin is connected to the inner circuit in a different way
- We just have to know how to use each pin!
- Contains several components
  - Comparators, Flip Flops, Transistors



What's in the 555 Timer



OPEN PROJECT SPACE

IEEE

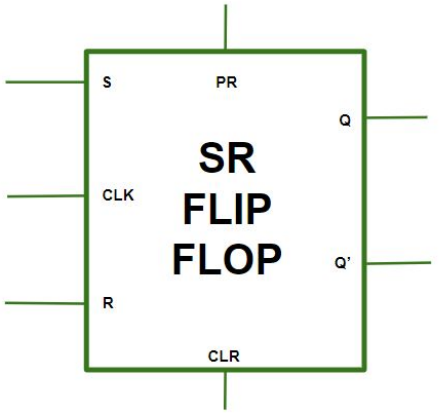
IEEE UF Student Branch

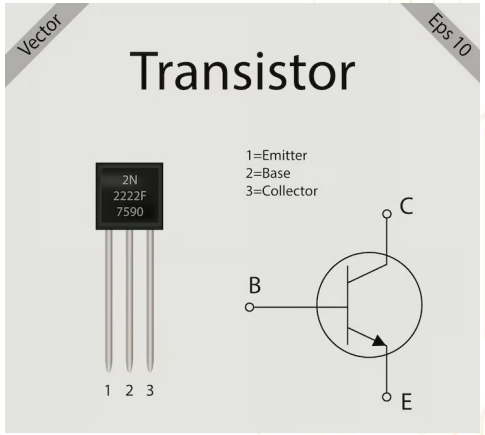
Components

OPEN PROJECT SPACE

Comparators, transistors, SR Flip Flop



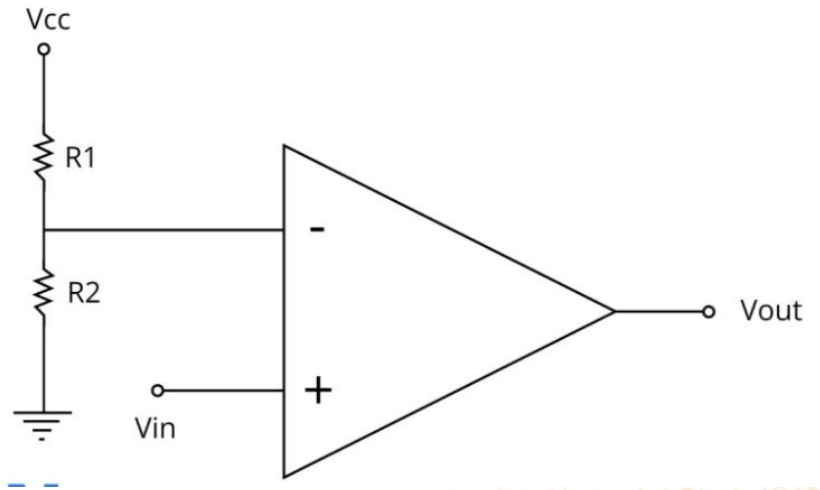




IEEE
IEEE UF Student Branch

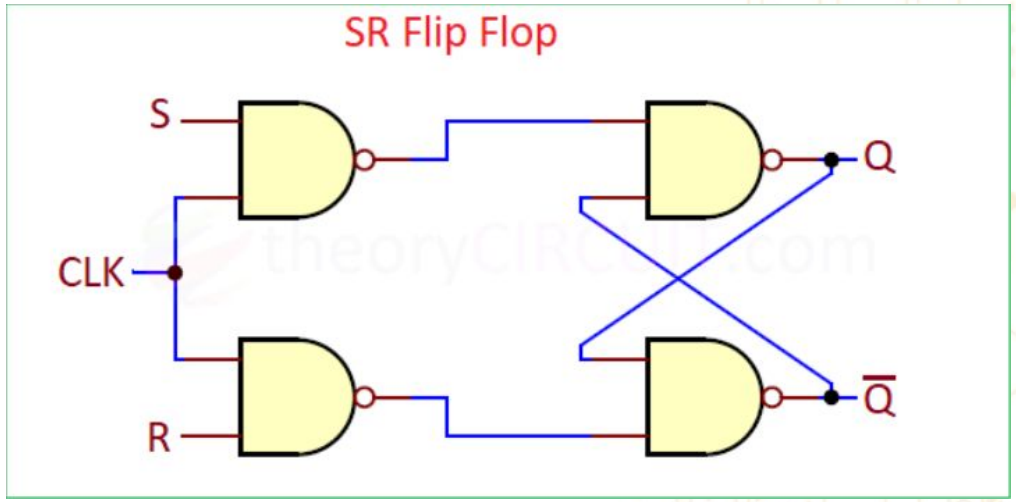
Comparator

- Evaluates 2 input voltages, making a binary decision
- Non-inverting input
  - The sensing pin that is triggered when higher than the reference
- Inverting input (Reference Pin)
  - Sets baseline for decisions
- Output Pin
  - Delivers the digital result (Vcc or GND) to the next stage of the circuit



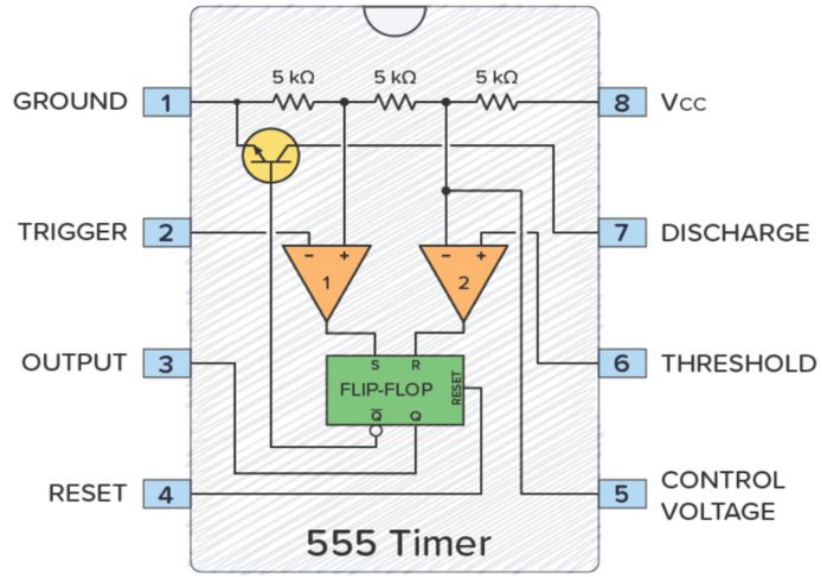
SR Flip Flop

- Stores one bit of binary data
- Acts as the memory of the circuit
  - Remembers the binary data even after the initial input disappears
- There are multiple types of Flip Flops!
  - D-FF, T-FF, SR-FF, JK-FF



Discharge Transistor

- Connected to pin 7 of the 555 timer
- Functions to rapidly drain the charge from the capacitor
  - Acts as a reset to the timing loop
- When the SR-FF resets, the transistor turns on -> draining the capacitor instantly



OPEN PROJECT SPACE

Introducing the 8-bit Shift Register

IEEE UF Student Branch

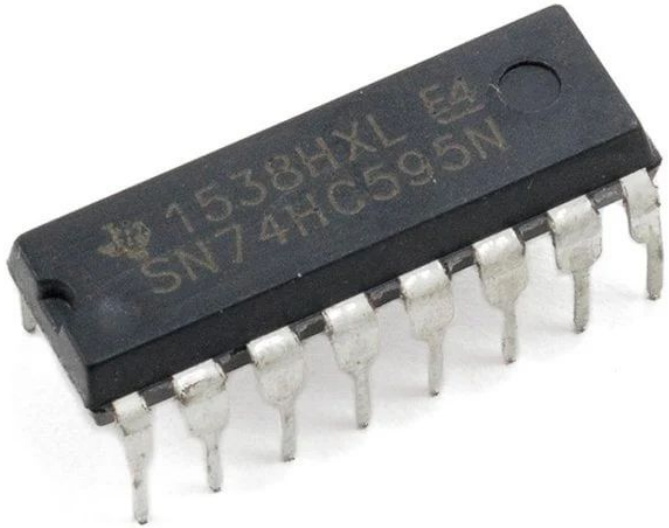
The Goal

- Using an 8-bit shift register, make a visual sequencer of each frequency from your 555-timer piano using 4 LEDs
    - Each LED should light up one after the other
- The shift register's input is connected to the 555-timer's output

8-bit Shift Register (74HC595)

OPEN PROJECT SPACE

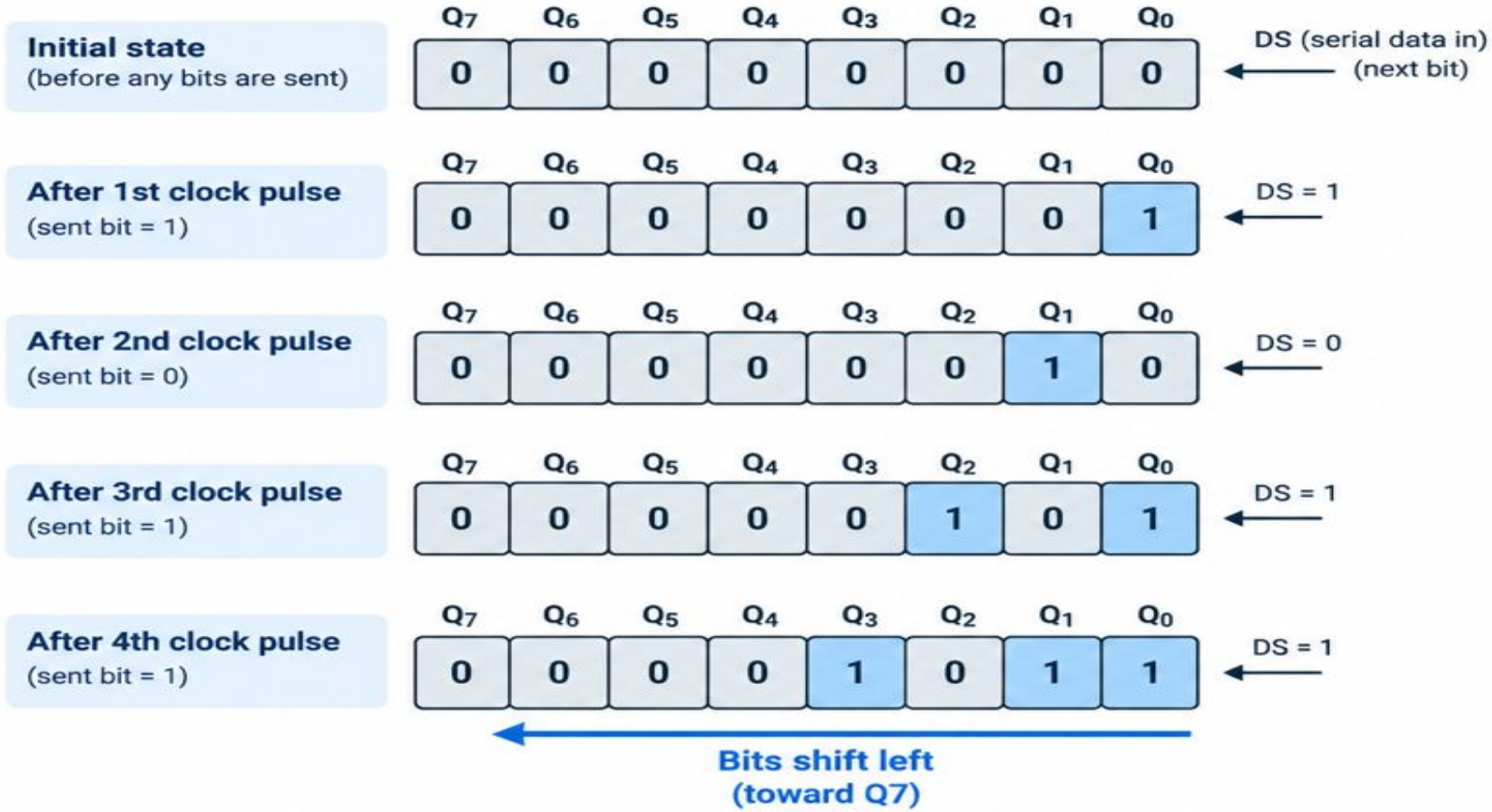
- Purpose: Controls 8 outputs with only 3 MCU pins
- Functionality
  - Bits enter the data pins serially with a clock pulse pushing each bit into the next register spot
  - A latch signal is used to prevent flickering in the output



IEEE
IEEE UF Student Branch

Visualization of 74HC595

OPEN PROJECT SPACE

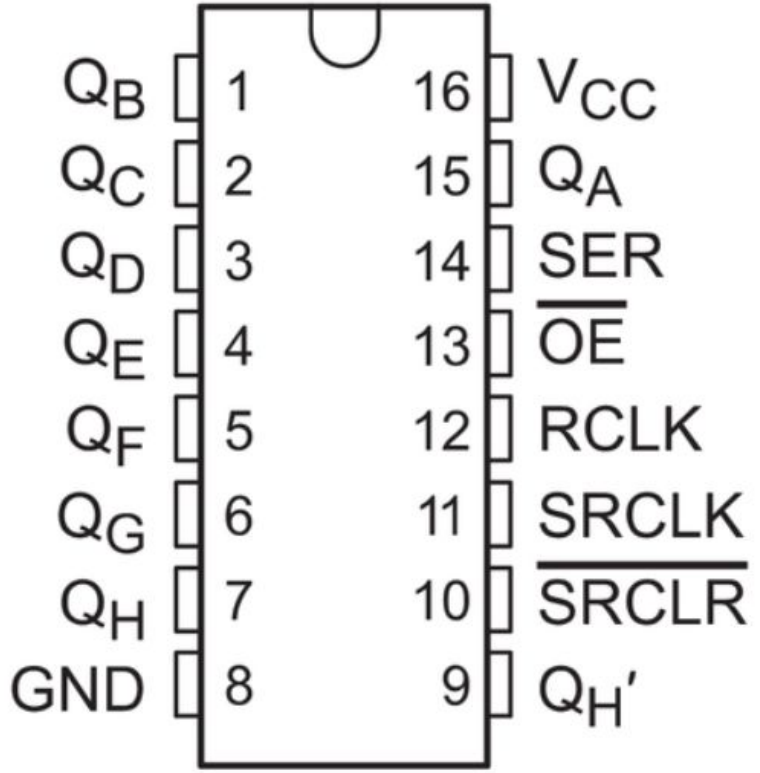


IEEE

IEEE UF Student Branch

MCU Pins

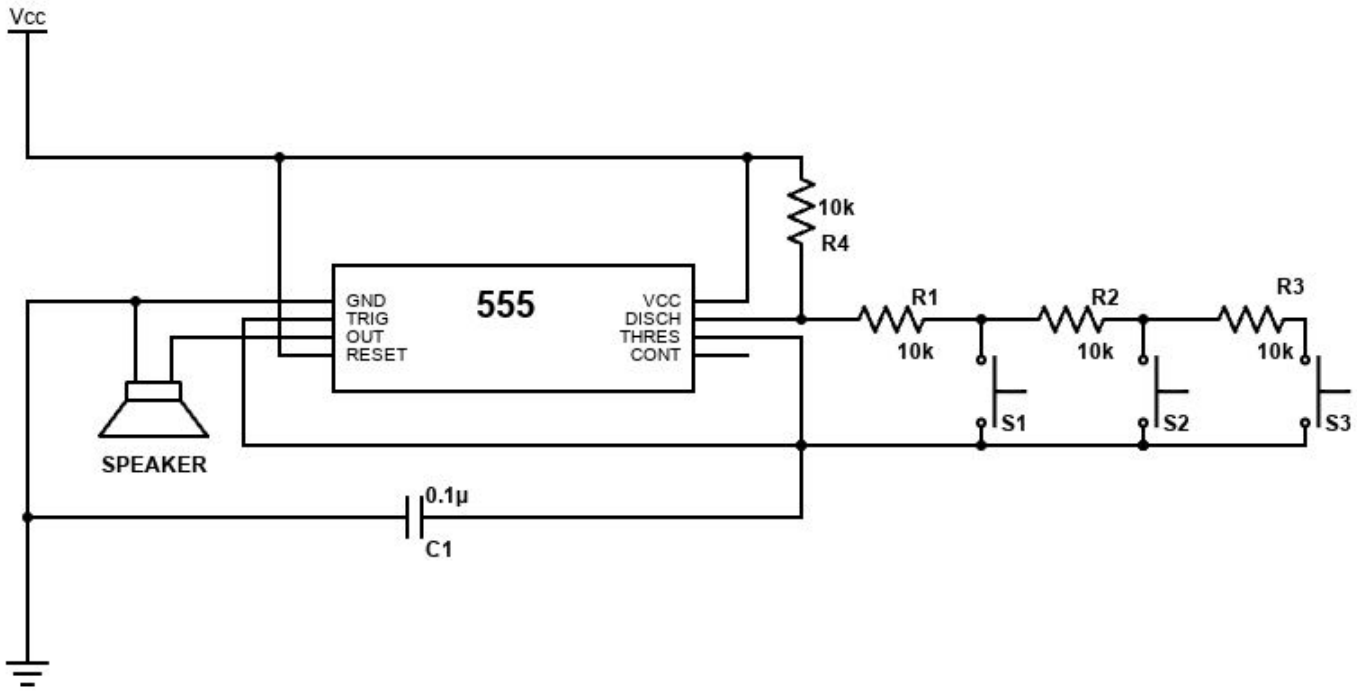
- SER (Data Pin): Sets each bit individually with a 1 or 0
- SRCLK (Shift Clock Pin): Uses the clock pulse to time when to look at the data pin, grab its value, and shift the entire chain of bits one to the right
- RCLK (Latch Clock Pin): Allows for all of the shifting to be done abstractly for 8 clock pulses until an output is presented



555 Timer Hints

OPEN PROJECT SPACE

Helpful schematic for 555 with speaker:

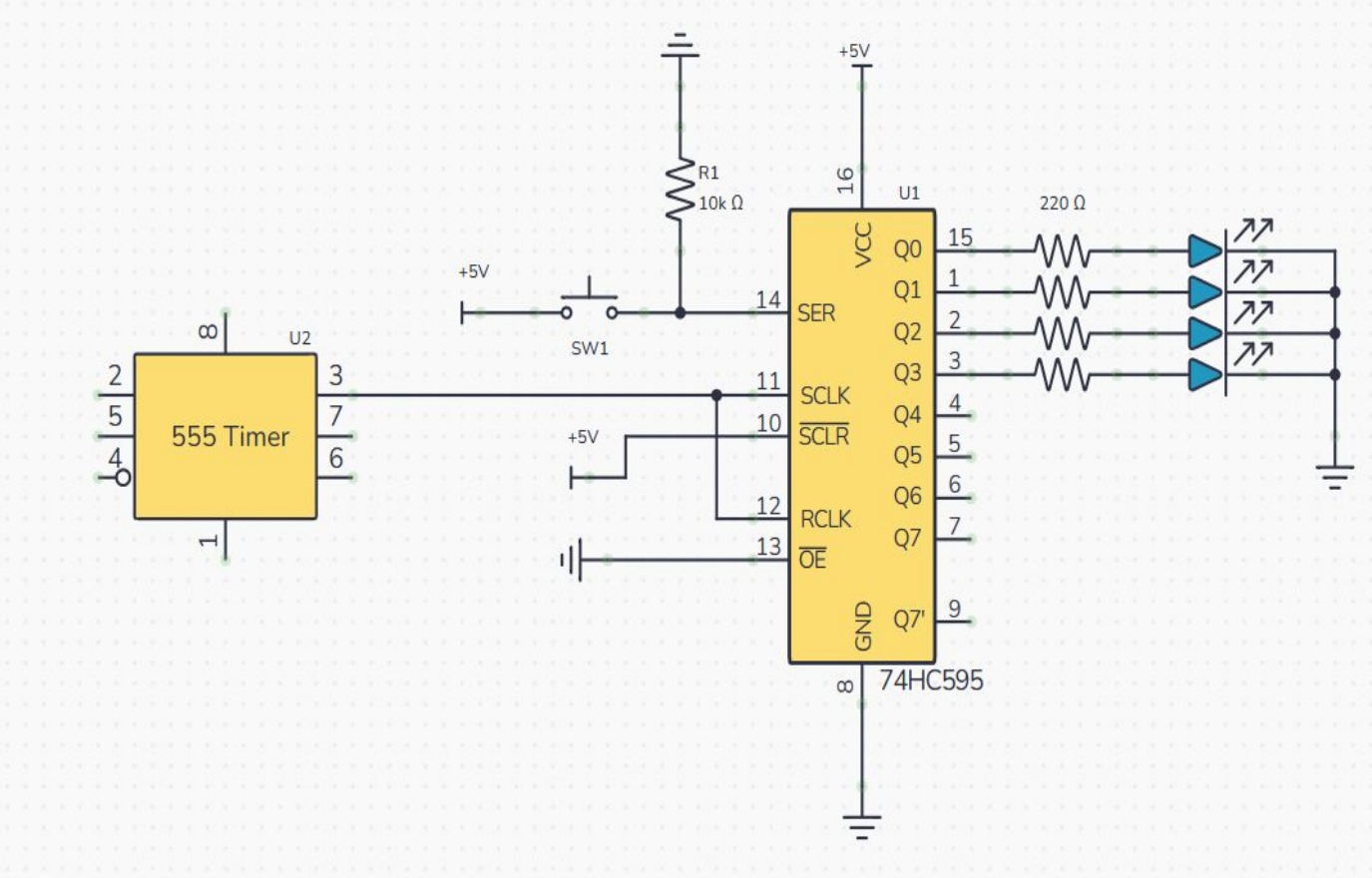


IEEE

IEEE UF Student Branch

8-bit Shift Register

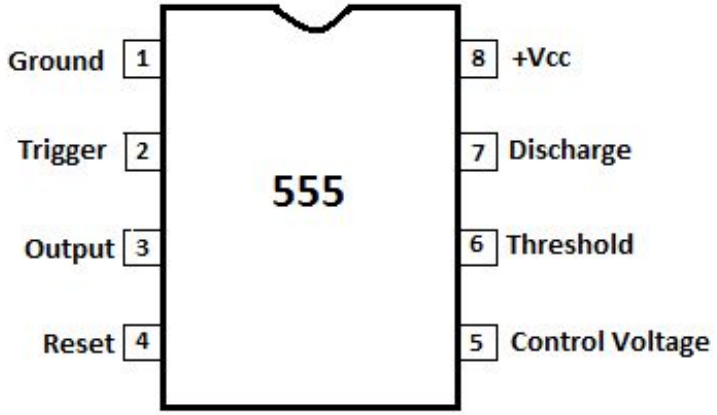
- Helpful schematic for adding 8 bit shift register
- Make sure to follow previous schematic for all 555 timer connections

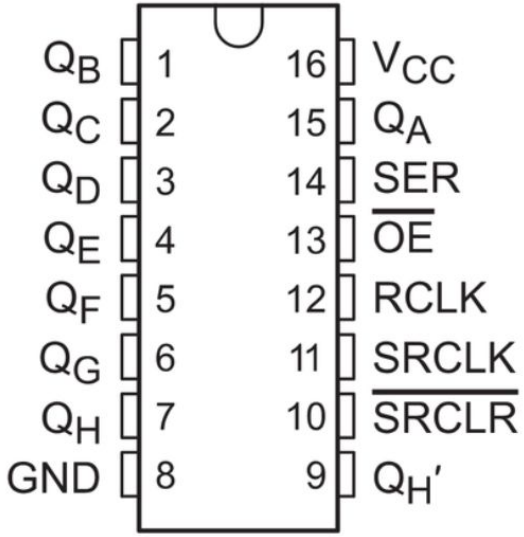


IEEE UF Student Branch

Pinouts

Remember 555 and 8-bit shift reg pinout diagram:

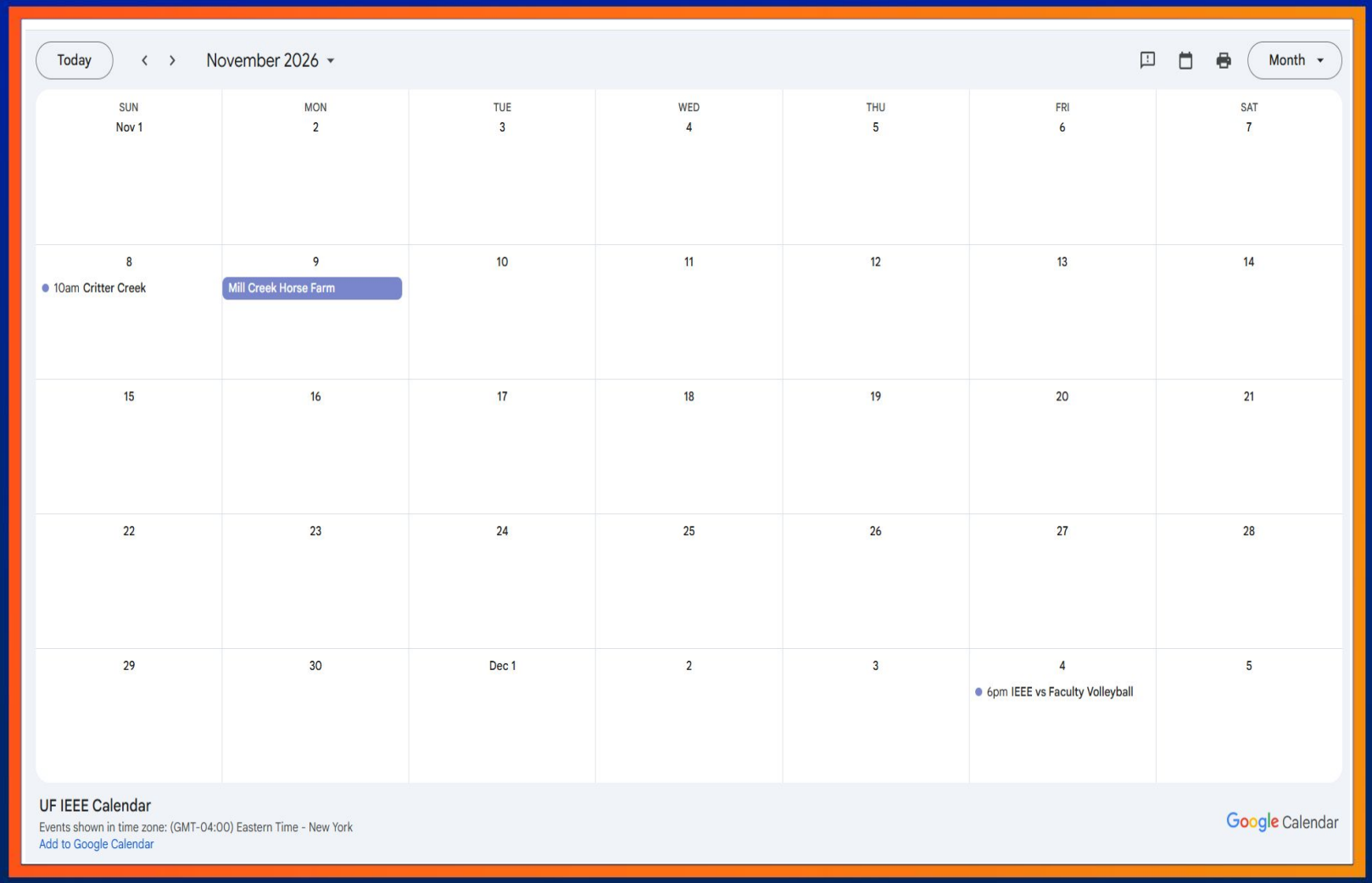




Good Luck

- Remember to attend office hours with any questions
- Refer to project 3 document

OPEN PROJECT SPACE



Keep up with IEEE!

Upcoming events:

11/8: Critter Creek (10 am)

11/9: Mill Creek Horse Farm

12/04: IEEE vs Faculty Volleyball

Reminders:

Project 3 due 11/10/26!

IEEE UF Student Branch